Script Documentation Deploy-CT-EDLP-EgressControl.ps1

Summary

Deploy-CT-EDLP-EgressControl.ps1 is an interactive deployment utility for a Microsoft Purview egress-control portfolio. It defines 11 policies with one rule per policy. Ten policies target managed endpoints. One policy targets external Exchange email. All policies default to silent simulation unless the operator chooses another mode, and one mode is applied to every selected policy in a run.

The utility performs five main functions:

1. Collects the incident-report mailbox.
2. Connects to Microsoft Purview compliance PowerShell, installing or importing the required module first.
3. Lets the operator select policies and one deployment mode for all selected policies.
4. Retrieves active sensitivity labels and lets the operator select the trigger labels.
5. Resolves the sensitive service domain groups, creates or updates the selected policies and rules, then reports deployment status and outstanding actions.

The deployment architecture is shown below. It combines local operator interaction, compliance PowerShell, sensitivity-label discovery, service domain group resolution, policy creation or update, Activity Explorer validation, and incident-mailbox reporting.

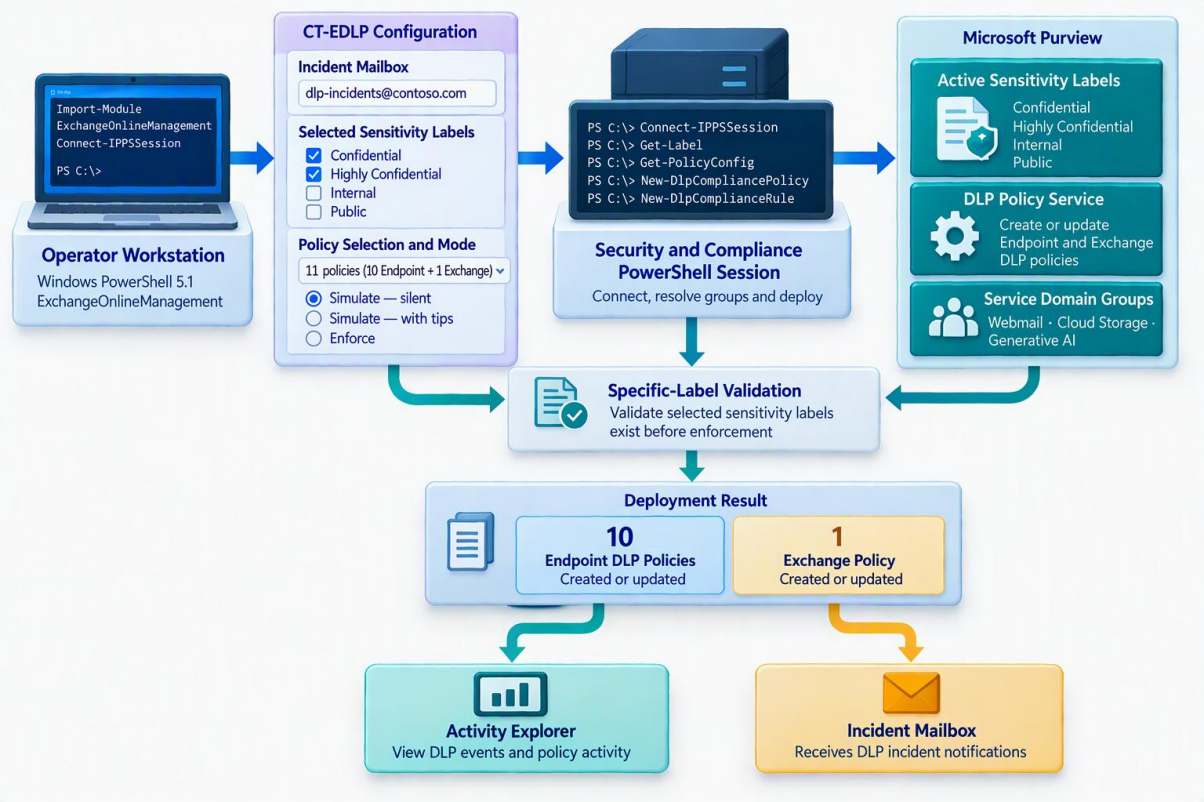


Figure 1: Operator interaction and deployment architecture for the CT-EDLP portfolio, from local configuration through Microsoft Purview policy creation or update, service domain group resolution, simulation evidence, and incident reporting.

Purpose and Scope

The utility creates label-driven controls for these egress paths:

- Removable media
- Printing
- Clipboard
- Bluetooth
- Remote Desktop
- Network shares
- Personal webmail
- Unsanctioned cloud storage
- Generative AI websites
- Restricted applications
- External Exchange email

The controls are organized into four groups: Device Egress, Browser & Cloud Egress, Application Egress, and Communication Egress.

The resulting architecture places several independent controls around the same protected file. Each control evaluates a different transfer action or destination.

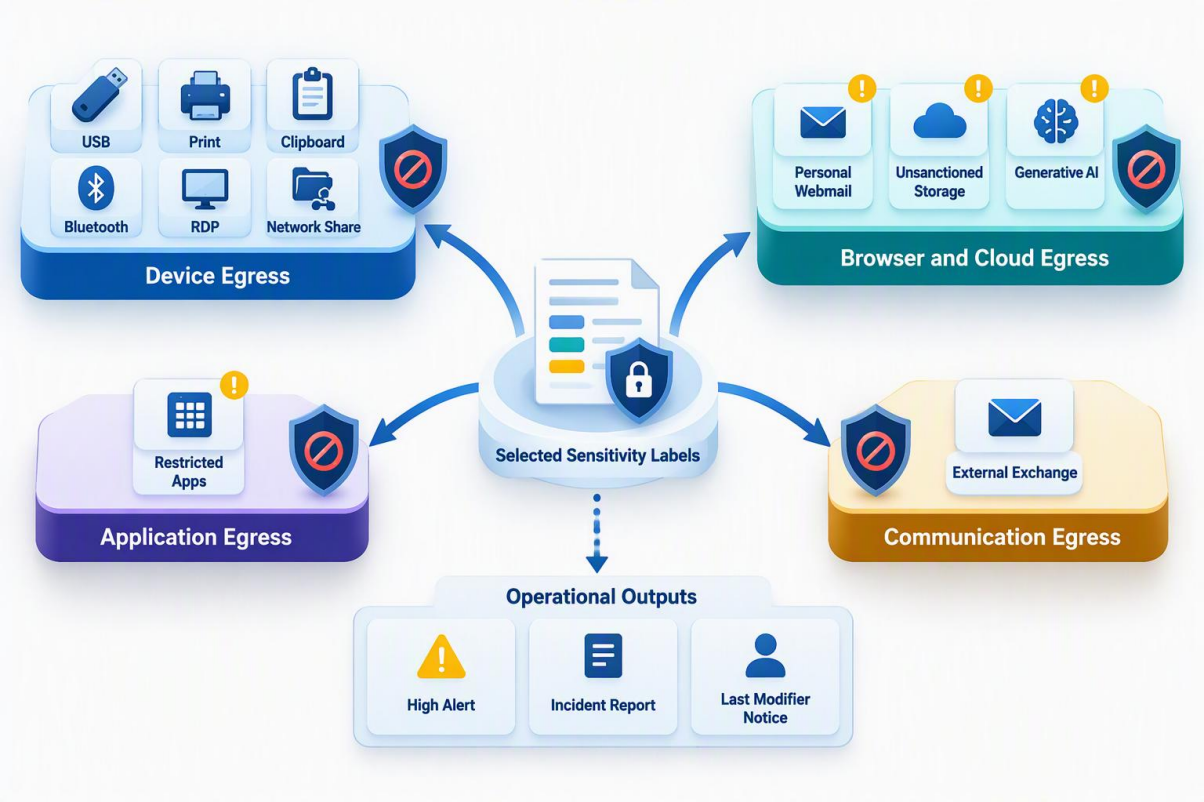


Figure 2: The resulting egress-control architecture combines device, browser and cloud, application, and communication controls around selected-label content.

Deployment Inventory

Group	Policy and Rule	Purview Location	Restriction or Action
Device Egress	CT-EDLP-01-RemovableMedia	All endpoint locations	RemovableMedia — Block
Device Egress	CT-EDLP-02-Print	All endpoint locations	Print — Block
Device Egress	CT-EDLP-03-Clipboard	All endpoint locations	CopyPaste — Block
Device Egress	CT-EDLP-04-Bluetooth	All endpoint locations	UnallowedBluetoothTransferApps — Block
Device Egress	CT-EDLP-05-RDPCopy	All endpoint locations	RemoteDesktopServices — Block

Group	Policy and Rule	Purview Location	Restriction or Action
Device Egress	CT-EDLP-06-NetworkShare	All endpoint locations	NetworkShare — Block
Browser & Cloud	CT-EDLP-07-WebmailUpload	All endpoint locations	CloudEgress + PasteToBrowser — Block (webmail groups)
Browser & Cloud	CT-EDLP-08-CloudStorage	All endpoint locations	CloudEgress + PasteToBrowser — Block (cloud storage groups)
Browser & Cloud	CT-EDLP-09-GenerativeAI	All endpoint locations	AccessByAIAgent + CloudEgress + PasteToBrowser — Block (Generative AI Websites group)
Application Egress	CT-EDLP-10-UnallowedApps	All endpoint locations	UnallowedApps — Block
Communication Egress	CT-EDLP-11-Exchange	All Exchange locations	External access — Block

The policy and rule names are identical. Each rule uses priority 0. Each configured rule reports High severity, generates an alert and incident report, and targets the last modifier for notification. CT-EDLP-07, CT-EDLP-08, and CT-EDLP-09 scope their CloudEgress and PasteToBrowser blocks to sensitive service domain groups when those groups can be resolved in the tenant.

Prerequisites

Operator Workstation

The execution host requires:

- Windows PowerShell 5.1.
- An interactive Windows environment capable of displaying Windows Forms. Some dialog features rely on PowerShell 7 behavior; see Important Technical Caveats.
- Access to PowerShell Gallery if the Exchange Online module must be installed.
- Network access to Microsoft 365 compliance services.

PowerShell Module

The required module is ExchangeOnlineManagement. If it is unavailable, the utility installs it for the current user with force and allow-clobber options. It then imports the module and displays the loaded version.

Administrative Access

The header requires Microsoft Purview compliance portal administrative rights. If the DLP commands are unavailable after connection, the reported possible causes are:

- Missing Compliance Administrator or Global Administrator role.
- Missing Microsoft Purview Compliance licensing.
- An outdated ExchangeOnlineManagement module.

For production administration, Microsoft recommends using the least-privileged role that fulfills the task rather than routinely using Global Administrator.

Permissions in the Microsoft Purview portal

Endpoint Readiness

Microsoft's Endpoint DLP guidance requires devices to be onboarded and reporting to Activity Explorer before endpoint scenarios are validated.

Endpoint DLP policy scenarios in Microsoft Purview

Licensing must be checked against the applicable Microsoft 365 and Purview service plans. Microsoft's service description states that Endpoint DLP evaluates policies for scoped users on onboarded devices.

Tenant Configuration Prerequisites

The script depends on two companion scripts that must have been run successfully before this utility: Add-WebmailDomainsToDlpGroups.ps1 and Add-CloudStorageDomainsToDlpGroups.ps1. They create the sensitive service domain groups required by CT-EDLP-07 (webmail) and CT-EDLP-08 (cloud storage). Without these groups, the CloudEgress and PasteToBrowser blocks of both rules fall back to unscoped blocks.

Before enforcement, Contoso must also configure:

- Sensitive service domain groups for personal webmail (Webmail - Global & Privacy and Webmail - Regional), created by running Add-WebmailDomainsToDlpGroups.ps1 beforehand (required for CT-EDLP-07).
- Sensitive service domain groups for unsanctioned storage (Cloud Storage - Global and Cloud Storage - Regional & Hosting), created by running Add-CloudStorageDomainsToDlpGroups.ps1 beforehand (required for CT-EDLP-08).
- AI capability and licensing for the generative AI control; the built-in Generative AI Websites group is referenced by its fixed identifier.
- Restricted App Groups for CT-EDLP-10.
- A monitored incident-report mailbox.
- Published sensitivity labels to select as triggers.

Operator Interaction

The operator completes three principal configuration interactions. The mailbox dialog opens before the compliance connection; the policy and label dialogs open after the connection succeeds.

Incident Mailbox

The first dialog requests the mailbox that receives DLP alerts and incident reports. The entered value must match a basic email-address format. Canceling ends execution without deployment.

Policy Selection and Mode

After the compliance connection is established, the policy dialog displays all four groups and their 11 policies. Every entry is selected by default. The operator can:

- Include or exclude entire groups.
- Include or exclude individual policies.
- Select one deployment mode that applies to all selected policies.
- Review a concise policy description.
- Select all or clear all policies, or cancel to end execution without deployment.

The modes, shown in the dialog as Simulate — silent, Simulate — with tips, and Enforce, are:

- TestWithoutNotifications: silent simulation.
- TestWithNotifications: simulation with policy tips.
- Enable: production block actions and notifications.

Sensitivity-Label Selection

After policy selection, the utility retrieves active labels. The selection dialog lists only leaf labels, meaning sublabels and top-level labels without sublabels, together with their parent label. Labels are ordered by parent priority and then child priority. The operator must select at least one label before continuing; canceling ends execution without deployment.

If the tenant returns no active labels, execution continues with an empty selection. Policies can then be created, but their rules are skipped and follow-up actions are recorded.

Execution Flow

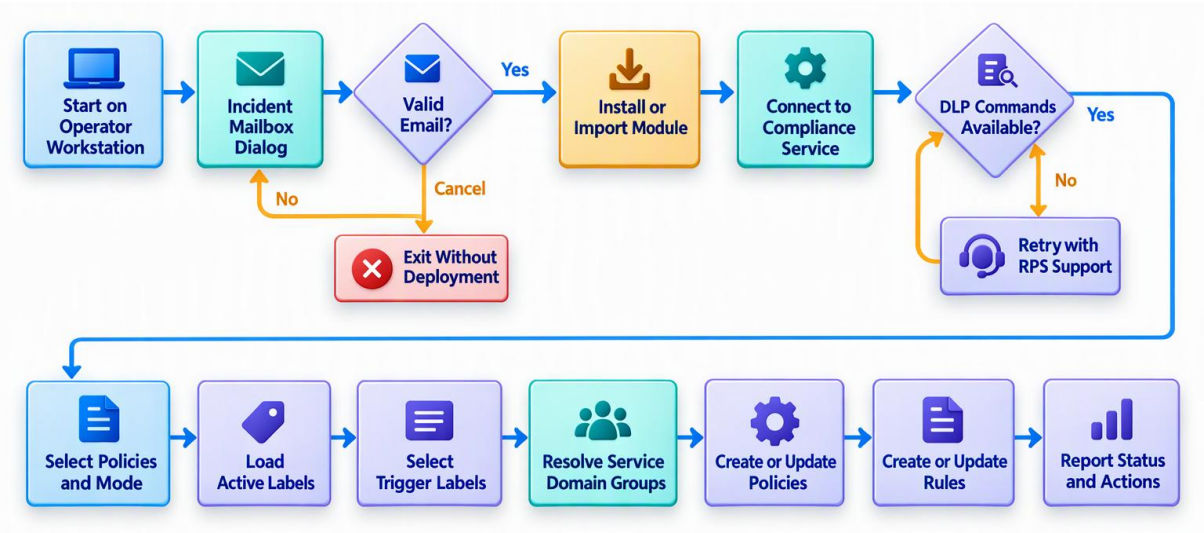


Figure 3: Execution flow for Deploy-CT-EDLP-EgressControl.ps1, from the incident mailbox and compliance connection through policy and label selection, service domain group resolution, policy and rule creation or update, and deployment reporting.

The initial compliance connection is retried after a failure. If the DLP commands remain unavailable, the utility attempts a connection with RPS support when the installed module exposes that option. Where supported, format-data loading is skipped to speed up the session. Execution stops if the commands are still unavailable. After connecting, the utility displays the tenant domain derived from the connection information.

Major Components and Responsibilities

Component	Responsibility
Show-MailboxDialog	Collects and validates the incident mailbox.
Show-LabelDialog	Displays active leaf labels with their parent label and returns their names and GUIDs.
Show-PolicyDialog	Collects policy inclusion and one deployment mode for all selected policies.
Connect-IPPS	Establishes the compliance PowerShell session, optionally requests RPS behavior, and skips format-data loading where supported.
New-LabelConditionJson	Builds the Exchange any-label and external-recipient condition.
New-EndpointLabelConditionJson	Builds the endpoint condition from the selected label GUIDs.
Get-SiteGroupMap	Reads the tenant policy configuration and maps sensitive service domain group names to their IDs.

Component	Responsibility
New-CloudEgressRestrictions	Builds CloudEgress and PasteToBrowser block restrictions scoped to the resolved groups, or unscoped blocks when none resolve.
Initialize-DlpPolicy	Checks selection, checks for an existing policy, updates its mode if needed, and creates a missing policy.
Initialize-DlpRule	Checks selection, updates an existing rule or creates a missing rule, and records failures.
Deployment summary	Queries all 11 expected policies and reports policy mode and rule count.

The three dialogs, connection function, condition builders, service domain group helpers, and initialization functions are explicitly separated in the execution design.

Common Rule Configuration

Every deployed rule receives a common operating baseline:

- The selected-label condition (selected label GUIDs for endpoint rules; any label for the Exchange rule).
- Last-modifier notification.
- The same bilingual policy-tip message.
- An alert to the configured incident mailbox.
- An incident report to the configured incident mailbox.
- High severity.
- Priority 0.
- A channel-specific blocking restriction, scoped to sensitive service domain groups for CT-EDLP-07 to CT-EDLP-09.

Incident reports request these fields:

- Title
- Service
- Matched item
- Matched rules
- Detections
- Document last modifier
- Sensitivity label
- Severity

CT-EDLP-11 additionally applies NotInOrganization, enables access blocking, and sets the block scope to All. Internal Exchange recipients are therefore outside its intended match scope.

Existing-Object Behavior

The utility now reconciles existing objects partially rather than only creating missing ones.

- If a policy already exists and its mode differs from the selected mode, the policy mode is updated; otherwise the policy is left unchanged.
- If its rule is missing, the rule is created under the existing policy.
- If a rule already exists, its actions, notifications, and comment are updated, followed by a separate update of its condition.
- Existing policy comments and locations are not updated.

This behavior makes reruns suitable for applying the current mode, conditions, and actions to existing objects. It is not a full drift correction, because policy comments, locations, and any additional rules are left untouched. An operator must record the existing configuration before a rerun, because updated settings overwrite the previous values.

Error Handling

The general error preference is Stop. The utility also maintains separate collections for pending actions, created policies, created rules, and skipped rules. A failed mode update on an existing policy produces a warning and the deployment continues.

Connection Failures

The first connection failure produces a warning and a retry. Missing DLP commands trigger an additional compatibility attempt. If the commands remain unavailable, execution terminates with role, license, and module-version guidance.

Policy Creation Failures

Policy creation is not wrapped in a catch block. An unhandled policy-creation error can therefore stop the deployment because the global error preference is Stop.

Rule Creation and Update Failures

Rule creation and rule update are handled per rule. A failed rule:

- Produces a warning.
- Is added to the skipped-rules list.
- Produces a manual recreation or update action.
- Allows the remaining deployment sequence to continue.

Missing Labels

A selected policy may be created without a rule when no label was selected or no active label was available. The summary must therefore be checked for policies showing zero rules.

Important Technical Caveats

Selected-Label Condition

The intended design targets the labels selected by the operator. The endpoint condition builder now places the selected label GUIDs in a ContentContainsSensitiveInformation condition, so CT-EDLP-01 to CT-EDLP-10 match only the selected labels. The Exchange condition builder still uses ContentIsNotLabeled = false, which means content has any label, and does not use the selected GUIDs.

Do not enable CT-EDLP-11 until its effective condition has been inspected and proven to match only the approved labels, and verify the endpoint conditions in the portal before enforcement.

PowerShell Version Compatibility

The script declares Windows PowerShell 5.1 as its minimum version, but two dialog details depend on PowerShell 7. The mailbox text box sets a placeholder property that the .NET Framework text box in Windows PowerShell 5.1 does not provide, and the policy tree uses a Unicode escape sequence that Windows PowerShell 5.1 does not interpret. Test the dialogs in the intended host before production use.

Portfolio-Description Wording

The introductory description calls all 11 objects Endpoint DLP policies. The deployed locations show that CT-EDLP-01 through CT-EDLP-10 are endpoint policies, while CT-EDLP-11 uses the Exchange location.

Dependency-Sensitive Controls

CT-EDLP-07 and CT-EDLP-08 use the same CloudEgress and PasteToBrowser restrictions but reference different sensitive service domain groups. Their separation depends on those groups existing; if they cannot be resolved, both rules fall back to an unscoped block. Paste-to-browser blocking does not trigger on label-matched content, so blocking paste to cloud sites requires a separate SIT-based rule. CT-EDLP-09 requires AI feature and licensing validation; its follow-up message still names the restriction GenerativeAIWebsites, while the deployed setting is AccessByAIAgent. CT-EDLP-10 requires Restricted App Groups.

Administrator Runbook

Before Execution

1. Confirm the approved change record and maintenance controls.
2. Confirm PowerShell 5.1 and interactive desktop access.
3. Confirm ExchangeOnlineManagement availability or approved installation access.
4. Confirm an appropriate Purview administrative role.
5. Confirm applicable Purview and Endpoint DLP licensing.
6. Confirm endpoints are onboarded and visible in Activity Explorer.
7. Confirm active sensitivity labels exist and that Add-WebmailDomainsToDlpGroups.ps1 and Add-CloudStorageDomainsToDlpGroups.ps1 have been run to create the sensitive service domain groups for CT-EDLP-07 and CT-EDLP-08.
8. Confirm the incident mailbox is monitored.
9. Record the existing state of identically named policies and rules.

During Execution

1. Enter the incident mailbox.
2. Authenticate to Microsoft 365 when prompted.
3. Select the required policies.
4. Keep the mode at Simulate — silent (TestWithoutNotifications) unless enforcement is separately approved.
5. Select the approved sensitivity labels.
6. Monitor warnings during group resolution and policy and rule creation or update.
7. Review the final policy status, mode, and rule count.
8. Capture all pending actions and skipped rules.

Post-Deployment Validation

Verify that:

- All expected policies exist.
- Every selected policy has exactly one rule.
- Policy modes match the mode selected for the run.
- Endpoint policies use the Devices location.
- CT-EDLP-11 uses Exchange.
- Effective label conditions are not broader than approved.
- High-severity alerts reach the incident mailbox.
- Incident reports contain the requested fields.
- Expected label events appear in Activity Explorer.
- Service domain groups and Restricted App Groups produce the intended differentiation.

Troubleshooting Guide

Symptom	Likely Source	Operator Action
Module cannot be loaded	Missing or outdated ExchangeOnlineManagement	Install or update the module, then rerun.
Compliance connection fails	Authentication or session issue	Review the warning and retry result.
DLP commands unavailable	Role, license, or module issue	Validate the three causes printed by the utility.
No labels appear	No active leaf labels returned	Confirm active published labels in Purview.
Policy exists but differs	Comments or locations are not updated	Compare and correct through an approved change process.
Policy has zero rules	No labels or rule-creation failure	Review skipped rules and pending actions.
Group not found warning, or webmail and storage behave identically	Service domain groups are missing or overlapping	Run Add-WebmailDomainsToDlpGroups.ps1 and Add-CloudStorageDomainsToDlpGroups.ps1 first to create the groups, correct any overlaps, then rerun.
AI rule fails or has no effect	AI capability or license not available	Validate AI Hub and DLP for AI Apps; confirm the Generative AI Websites group.
Restricted-app rule has no effect	Restricted App Groups are empty	Configure and test the required groups.
No incident email arrives	No matching event or mailbox issue	Confirm simulation events, mailbox address, and incident settings.

The rule-failure, domain, AI, app-group, and post-deployment actions are explicitly added to the pending-actions list for operator follow-up.

Rollback Considerations

No automated rollback workflow is defined. The deployment tracks created and skipped objects, but the final stage reports status and pending actions rather than reversing changes. Updated policy modes and rule settings are not recorded, so previous values must come from the pre-execution record.

A controlled rollback should therefore:

1. Identify which policies and rules were newly created.
2. Distinguish them from objects that existed before execution and restore any updated modes or rule settings from the pre-execution record.
3. Disable enforced policies before deletion when immediate risk reduction is required.
4. Remove only approved newly created rules and policies.
5. Preserve incident and Activity Explorer evidence required for review.
6. Revalidate the remaining DLP estate after rollback.

Care is required when a policy existed before the run but its rule was added during the run. In that case, rollback should target the newly added rule rather than the preexisting policy.